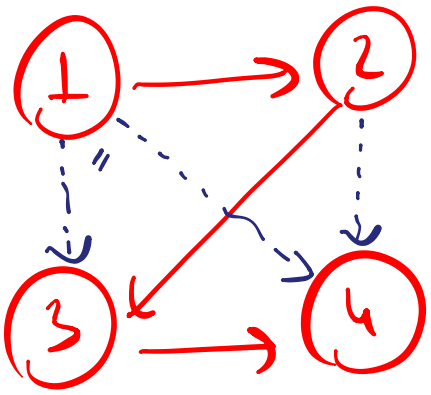
8.4

1. Apply Warshall's algorithm to find the transitive closure of the digraph defined by the following adjacency matrix:

$$R^{(0)} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R^{(k)}[i,j] = \begin{cases} R^{(k-1)}[i,j] \\ R^{(k-1)}[i,k] \text{ or } R^{(k-1)}[k,j] \end{cases}$$



$$R^{(1)} = R^{(0)} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$1 \to 2$

$2 \to 3$

$1 \xrightarrow{2} 3$

$$R^{(2)} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$1 - 3 \to 4$

$2 - 3 \to 4$

$$R^{(3)} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

4

1 - 4

$$R^{(4)} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \text{Final}$$

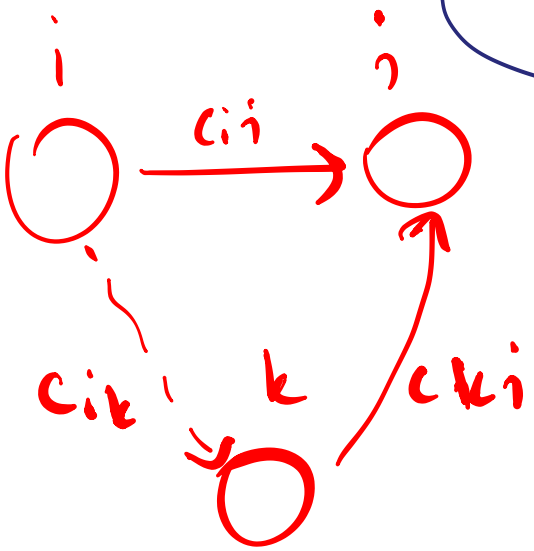
8.4

7. Solve the all-pairs shortest-path problem for the digraph with the following weight matrix:

$D^{(0)} =$

0	2	∞	1	8
6	0	3	2	∞
∞	∞	0	4	∞
∞	∞	2	0	3
3	∞	∞	∞	0

$$D^{(k)}[i,j] = \min\{D^{(k-1)}[i,j],\ D^{(k-1)}[i,k] + D^{(k-1)}[k,j]\}$$



$D^{(1)} =$

0	2	∞	1	8
6	0	3	2	14
∞	∞	0	4	∞
∞	2	2	0	3
3	5	∞	4	0

$$D^{(2)} = \begin{bmatrix} 0 & 2 & 5 & 1 & 8 \\ 6 & 0 & 3 & 2 & 14 \\ \infty & \infty & 0 & 4 & \infty \\ \infty & \infty & 2 & 0 & 3 \\ 3 & 5 & 8 & 4 & 0 \end{bmatrix}$$

$$D^{(3)} = \begin{bmatrix} 0 & 2 & 5 & 1 & 8 \\ 6 & 0 & 3 & 2 & 14 \\ \infty & \infty & 0 & 4 & \infty \\ \infty & \infty & 2 & 0 & 3 \\ 3 & 5 & 8 & 4 & 0 \end{bmatrix}$$

$$D^{(4)} = \begin{bmatrix} 0 & 2 & 3 & 1 & 4 \\ 6 & 0 & 3 & 2 & 5 \\ \infty & \infty & 0 & 4 & 7 \\ \infty & \infty & 2 & 0 & 3 \\ 3 & 5 & 6 & 4 & 0 \end{bmatrix}$$

$$D^{(5)} = \begin{bmatrix} 0 & 2 & 3 & 1 & 4 \\ 6 & 0 & 3 & 2 & 5 \\ 10 & 12 & 0 & 4 & 7 \\ 6 & 8 & 2 & 0 & 3 \\ 3 & 5 & 6 & 4 & 0 \end{bmatrix}$$

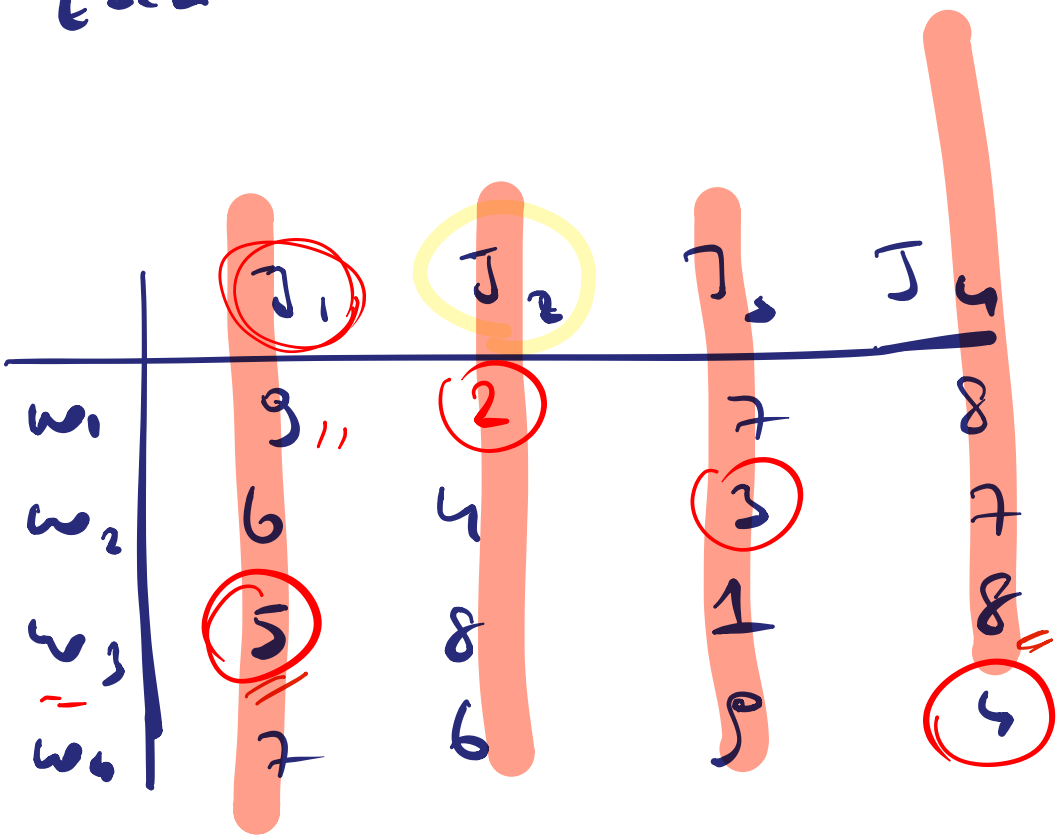
9.1

2. Design a greedy algorithm for the assignment problem (see Section 3.4). Does your greedy algorithm always yield an optimal solution?

n Jobs, n workers

→ Each job can be assigned only one worker.

→ Each worker can perform only one Job

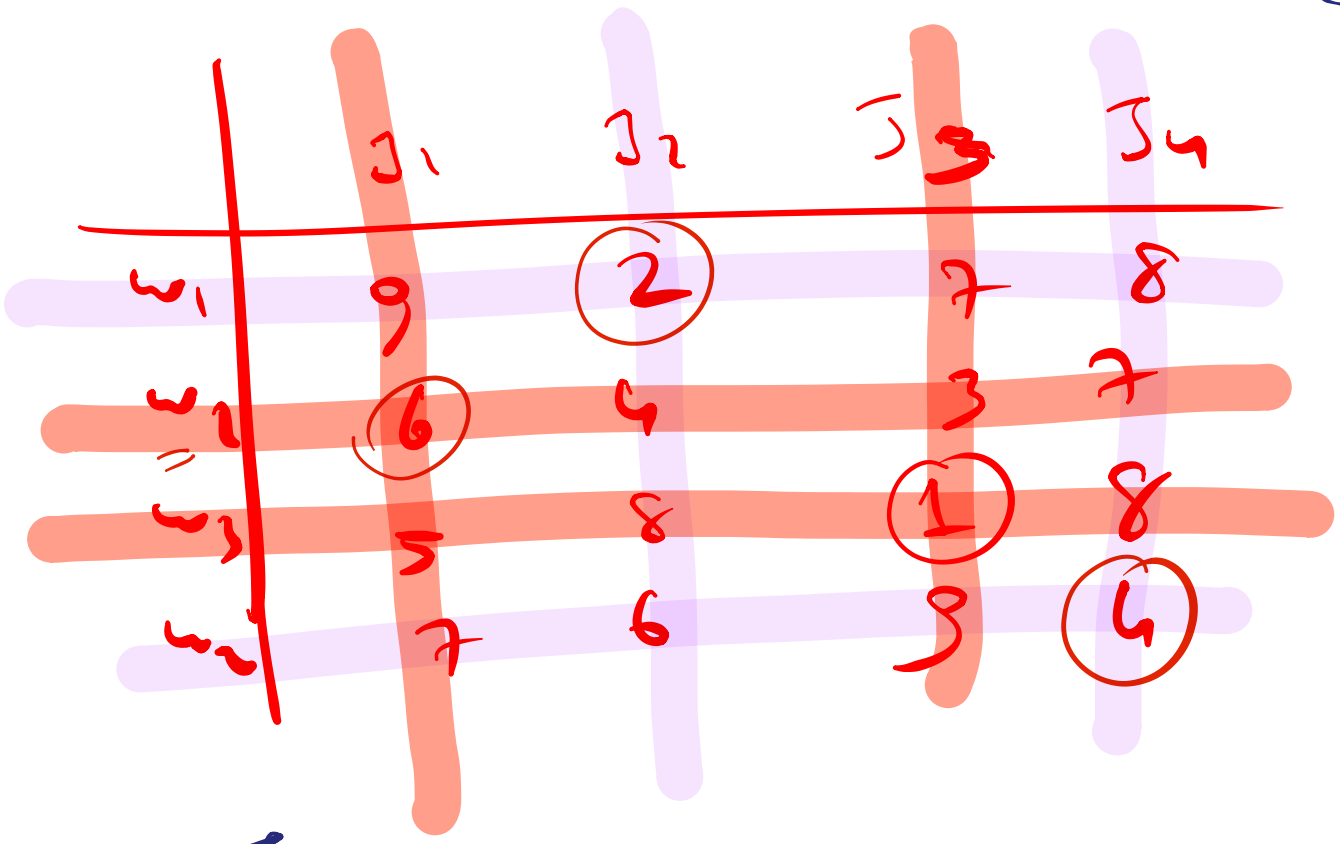


1st approach (Row by Row)

- Select the smallest value in each row, which is not in a previously marked column.
- Then marked its column.

Total cost = 2 + 3 + 5 + 4 = 14

→ 2nd way: Select smallest element in the unmarked rows and columns of the cost matrix, then marks its row and column



W1 – J2
W2 – J1
W3 – J3
W4 – J4

Total cost: 2 + 6 + 1 + 4 = 13 //

Does our greedy alg. always yield an optimal solution?

NO

$$\begin{bmatrix} 1 & 2 \\ 2 & 5000 \end{bmatrix} \rightarrow$$

1st way: 1 + 5000

2nd way: 1 + 5000

optimal = 2 + 2

3. *Job scheduling* Consider the problem of scheduling n jobs of known durations $t_1, t_2, \ldots, t_n$ for execution by a single processor. The jobs can be executed the total time spent by all the jobs in the system. (The time spent by one job in the system is the sum of the time spent by this job in waiting plus the time spent on its execution.)

Design a greedy algorithm for this problem. Does the greedy algorithm always yield an optimal solution?

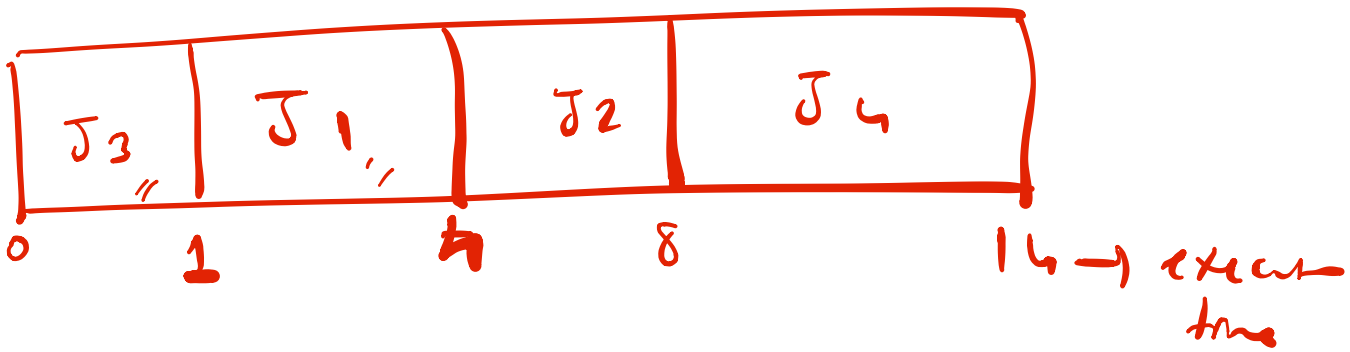
Total spent time = Execute time + waiting time

J	Execution Time
J_1	3
J_2	4
J_3	1
J_4	6

→ Sort jobs in increasing order of their execution times.

→ Then executes the jobs in that order.

J_3, J_1, J_2, J_4



$T_{S1} = E_1 + w_1 = 3 + 1 = 4$ (E_1, E_3)

$T_{S2} = E_2 + w_2 = 4 + (E_3 + E_1) = 8$ (E_2, 4)

$T_{S3} = E_3 + w_3 = 1 + 0 = 1$ (E_3)

$T_{S4} = E_4 + w_4 = 6 + (E_1 + E_2 + E_3) = 14$ (E_4, 8)

$$T_{TOT} = T_{S1} + T_{S2} + T_{S3} + T_{S4}$$

$$= (E_1 + E_3) + (E_1 + E_2 + E_3) + (E_3) + (E_1 + E_2 + E_3 + E_4)$$

$$= 4 \cdot E_3 + 3 \cdot E_1 + 2 \cdot E_2 + 1 \cdot E_4$$

Smaller E's are assigned to larger numbers, so our alg. always yields OPTIMAL solution

→ To minimize such sum, we have to assign smaller E's to the larger numbers.

3.2

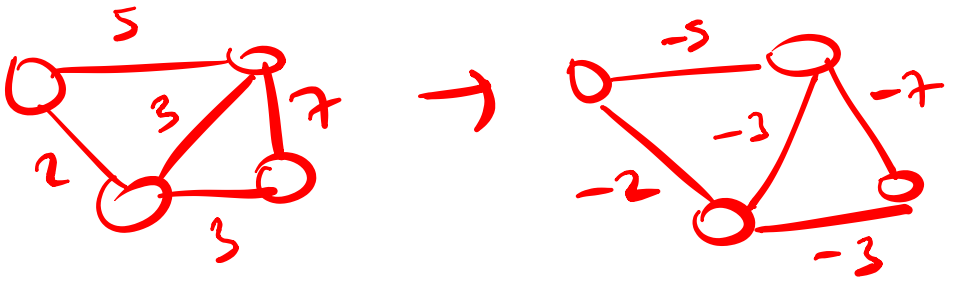
5. Design an algorithm for finding a ***maximum spanning tree***—a spanning tree with the largest possible edge weight—of a weighted connected graph.

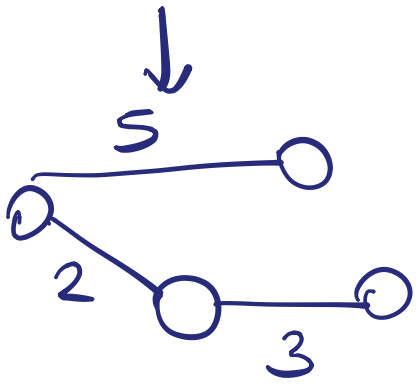
MST → Kruskal
→ Prim's

w(u,v) : weight of the edge (u,v)

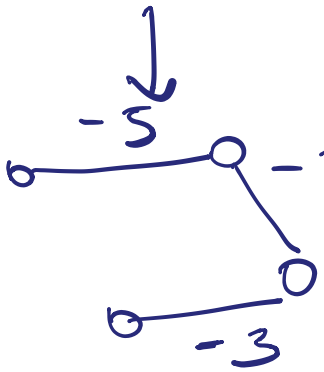
→ Replace w(u,v) with −w(u,v)

→ Then apply MST

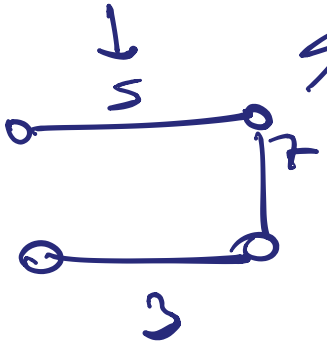




minimum spanning tree



minimum spanning tree



maximum spanning tree

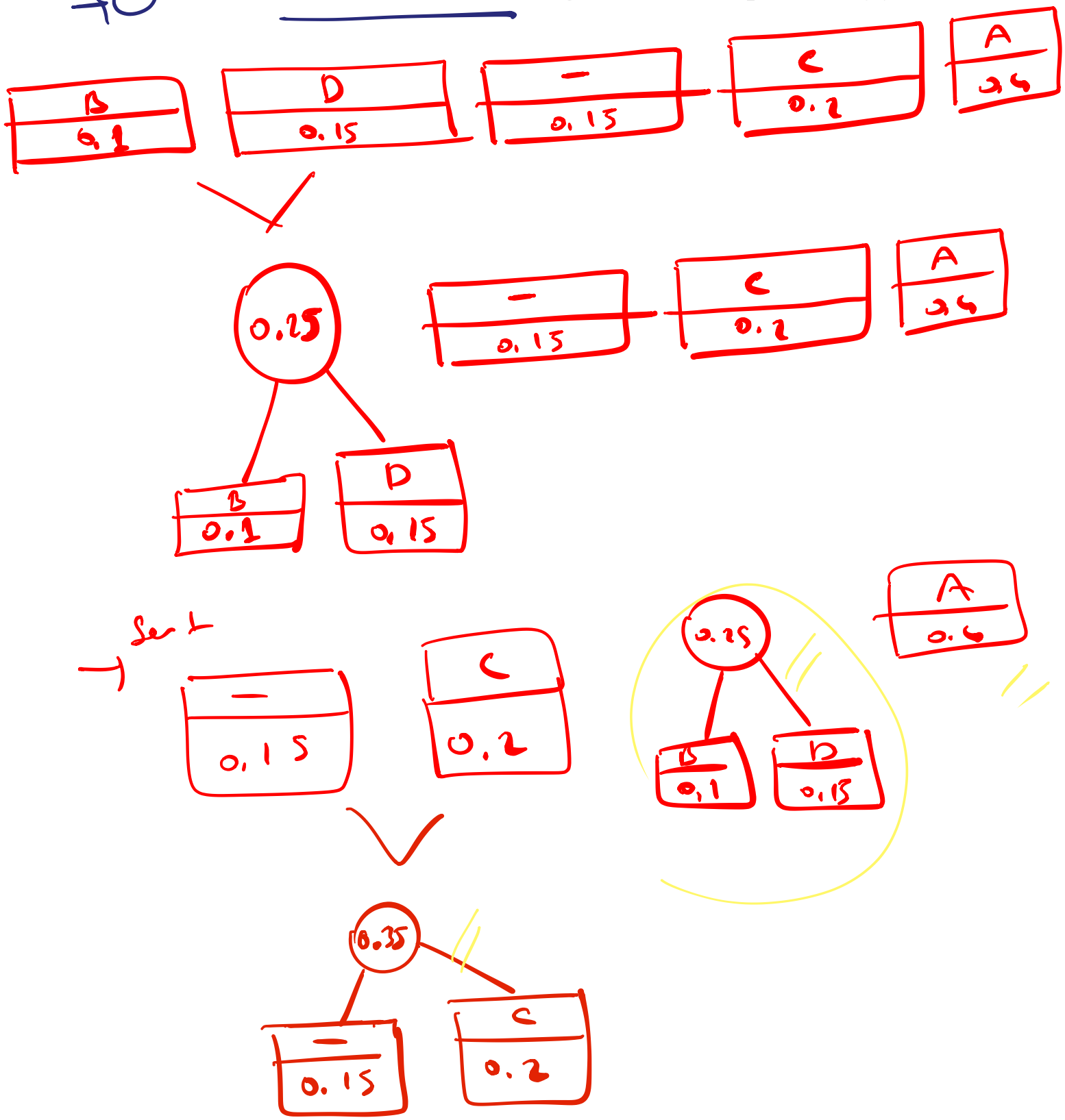
3.4

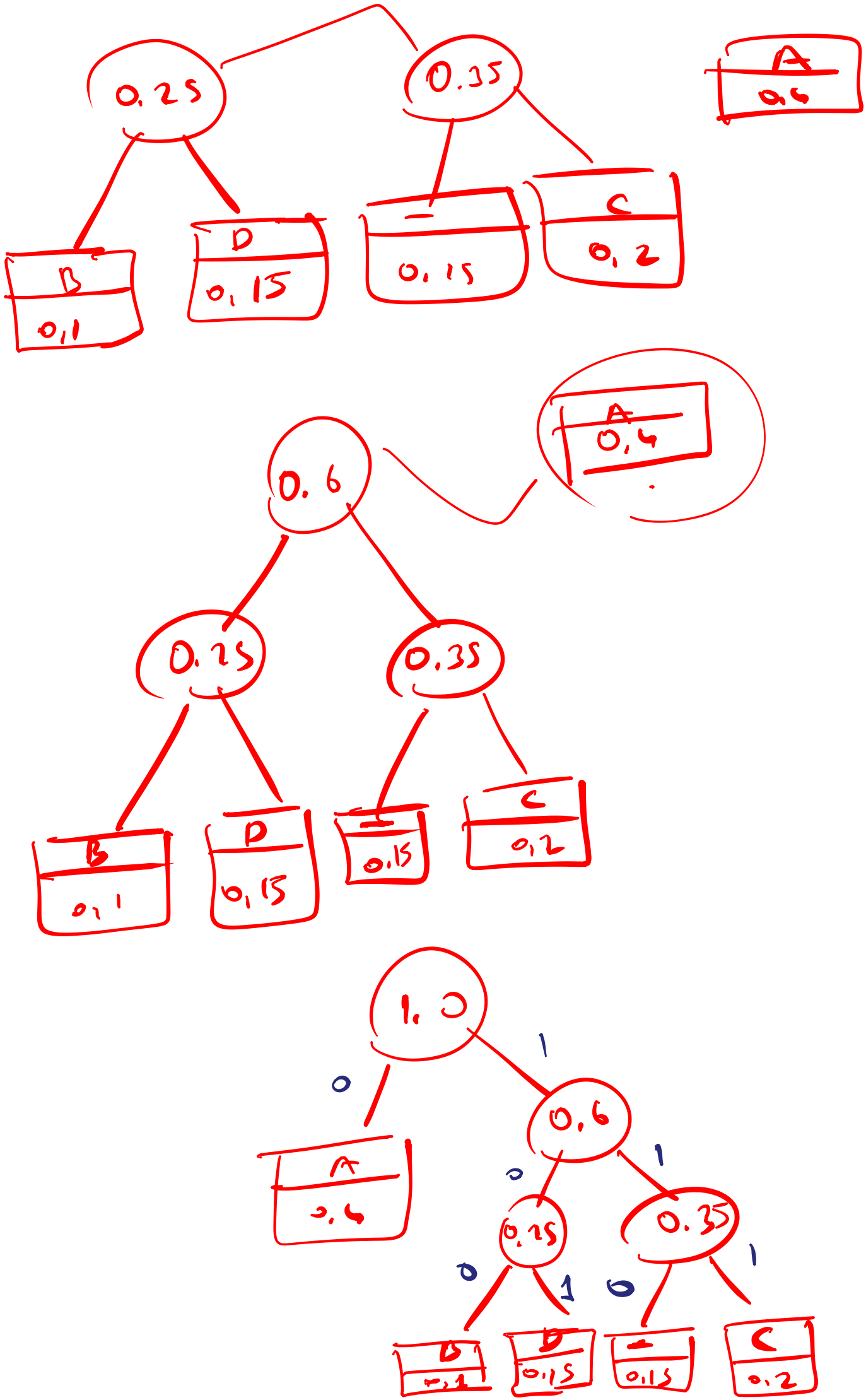
1. a. Construct a Huffman code for the following data:

symbol	A	B	C	D	_
frequency	0.4	0.1	0.2	0.15	0.15

b. Encode ABACABAD using the code of question (a).

c. Decode 100010111001010 using the code of question (a).

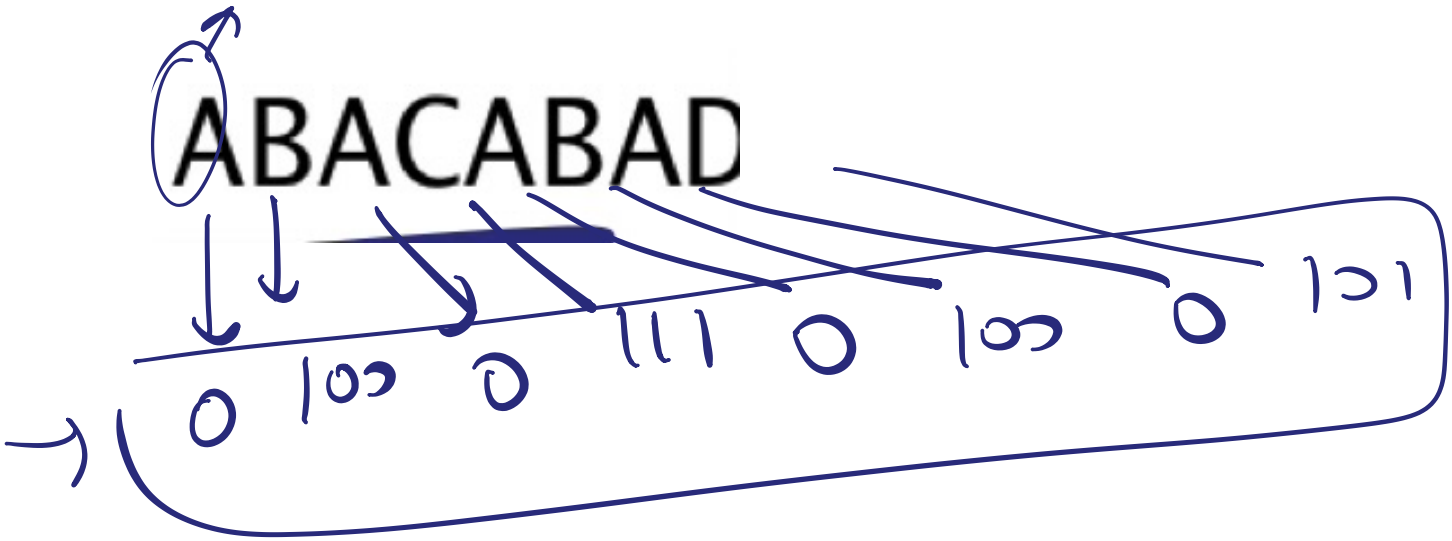



0.25
0.35
A
0.4
B
0,1
D
0,15
0,15
C
0,2
0.6
A
0,4
0.25
0.35
B
0,1
D
0,15
0.15
C
0,2
1.0
1
0
A
0,4
0,6
0
1
0,25
0.35
0
1
0
1
0,1
D
0,15
0,15
C
0,2

A	0
B	100
C	111
D	101
-	110

100010111001010

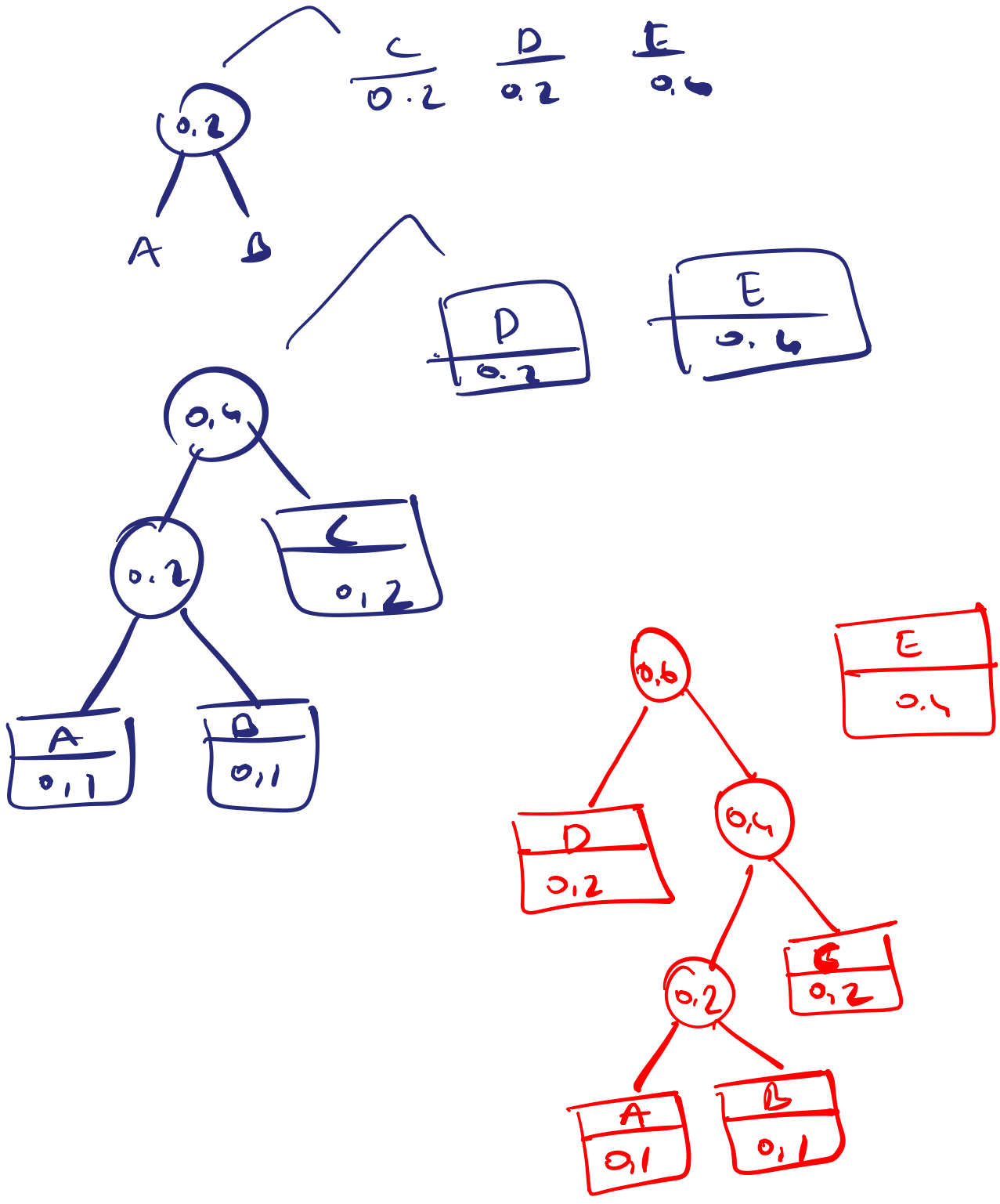
B A D - A B A

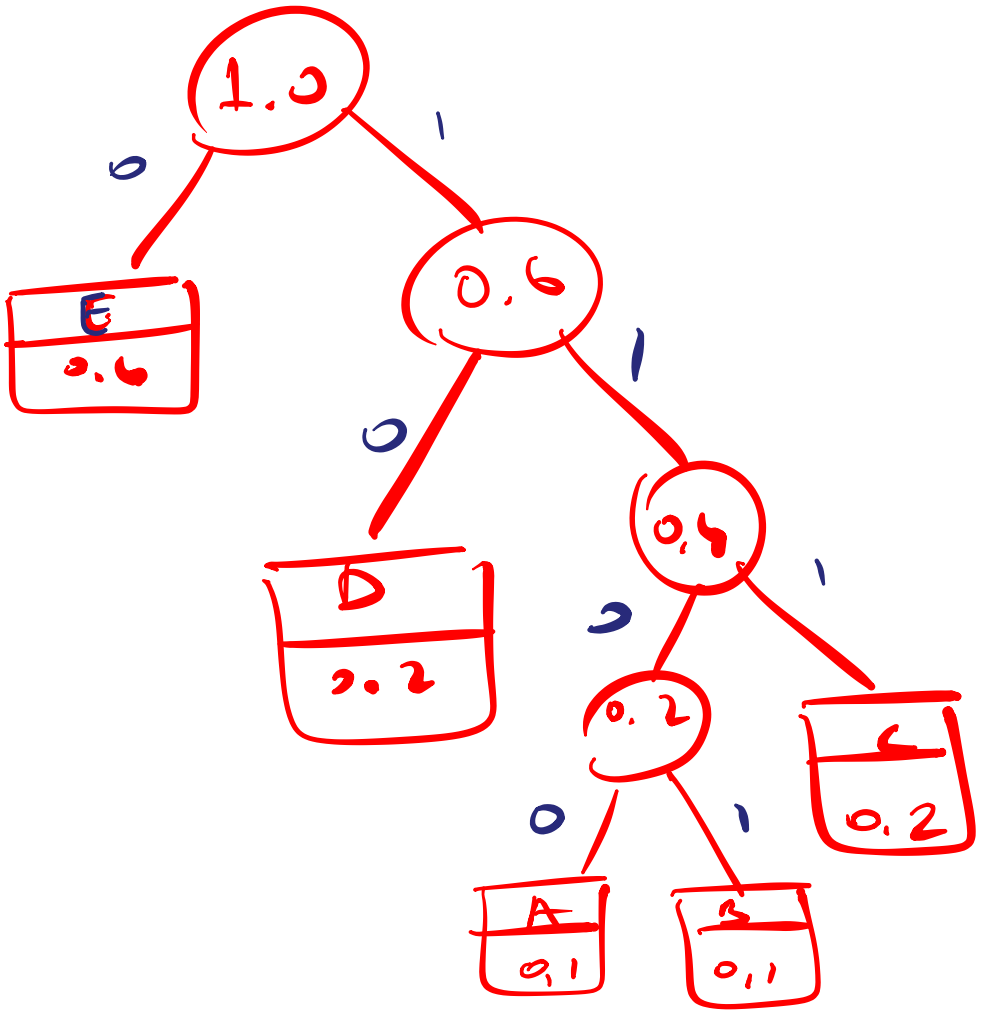


9.4

2. For data transmission purposes, it is often desirable to have a code with a minimum variance of the codeword lengths (among codes of the same average length). Compute the average and variance of the codeword length in two Huffman codes that result from a different tie breaking during a Huffman code construction for the following data:

symbol	A	B	C	D	E
probability	0.1	0.1	0.2	0.2	0.4





	Code	L
A	1101	4
B	1101	4
C	111	3
D	10	2
E	0	1

$$\bar{L} = 4.(0.1) + 4.(0.1) + 3(0.2) + 2(0.2) + 1.(0.4)$$

$$= 0.4 + 0.4 + 0.6 + 0.4 + 0.4 = 2.2$$

$$V = \sum_{i=1}^{n} (x_i - \bar{x})^2 p_i$$

$$= (4 - 2.2)^2 \times 0.1 + (4 - 2.2)^2 \times 0.1 +$$

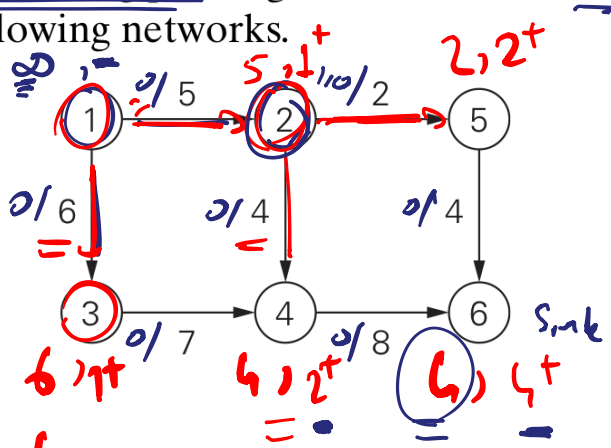
$$(3 - 2.2)^2 \times 0.2 + (2 - 2.2)^2 \times 0.2 +$$

$$(1 - 2.2)^2 \times 0.4 = 1.36$$

10.2

2. Apply the shortest-augmenting path algorithm to find a maximum flow and a minimum cut in the following networks.

a.



X_{ij} = used capacity

U_{ij} = edge capacity

r_{ij} = remaining capacity

BFS

Q: 1 2 3 4 5 6

$l_i = \min\{r_{ij}, l_j\}$

Augmenting path : $1 \to 2 \to 4 \to 6$

$l_2 = \min\{\infty, 5\} = 5$

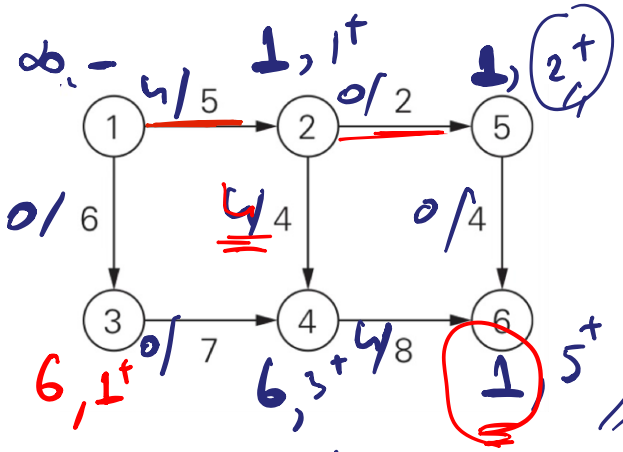
Augmenting Flow = 4

$l_3 = \min\{\infty, 6\}$

$l_4 = \min\{l_2, r_{24}\} = \min\{5, 4\}$

$l_5 = \min\{l_2, r_{25}\} = \min\{5, 2\}$

$l_6 = \min\{l_4, r_{46}\} = \min\{4, 8\}$



$l_7 = \min\{l_1, r_{12}\} = 1$

$\infty \quad 1, 1$

$u_{12} - x_{12}$

$5 \quad 4$

$= 1$

$\theta: \not{1}\ (2)\ \not{3}\ 5\ 4$

$l_3 = \min\{l_1, r_{13}\}$

$\infty \quad 6$

Augmenting path:

$1 \to 2 \to 5 \to 6$

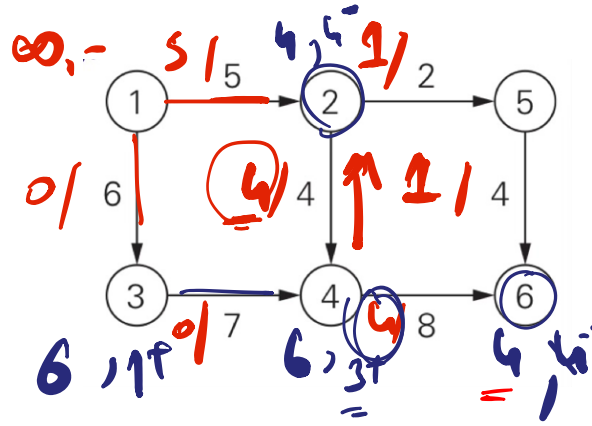
$l_5 = \min\{\underset{1}{l_2}, \underset{2}{r_{25}}\} = 1$

Augment by 1

$l_4 = \min\{\underset{4}{l_3}, \underset{7}{r_{34}}\}$

$l_6 = \min\{l_5, r_{56}\}$

$1 \quad 4$



$l_3 = \min\{l_1, r_{13}\}$ = 4, 6

$l_4 = \min\{l_3, r_{34}\}$ = 6, 7

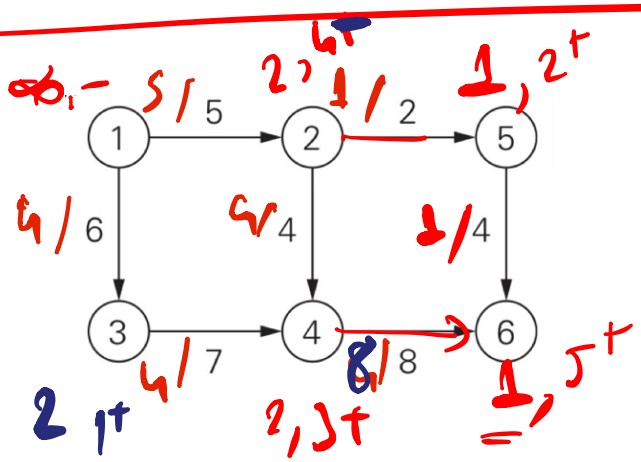
$l_2 = \min\{l_4, x_{42}\}$ = 6, 4

$l_6 = \min\{l_4, r_{46}\}$ = 6, 4

Q: 1 3 4 2 6

A-prog path : 1 → 3 → 4 → 6

flow = 4



$l_3 = \min\{l_1, r_{13}\}$ = ∞, 2

$l_4 = \min\{l_3, r_{34}\}$ = 2, 3

$l_2 = \min\{l_4, x_{42}\}$ = 2, 4

$l_5 = \min\{l_2, r_{25}\}$

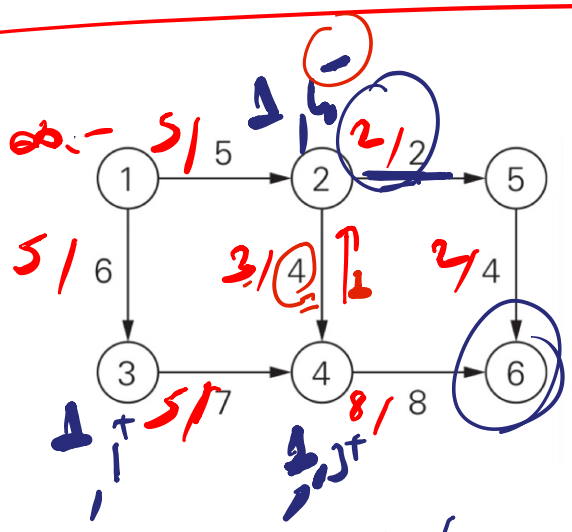
Q: 1 3 4 2 5

1 → 3 → 4 ← 2 → 5 → 6

flow = 1

$l_2 = 2$ $l_1 = 1$

$l_6 = \min\{l_5, c_{56}\}$
$1 \quad 3$



$l_3 = \min\{l_1, c_{13}\}$
$\infty \quad 1$

$l_4 = \min\{l_3, c_{34}\}$
$1 \quad 2$

$l_2 = \min\{l_1, x_{42}\}$
$1 \quad 3$

$2 + 8 = 10$ → max flow

min-cut: (2,5), (4,6)

(y from labeled nods to unlabeled nodes)